

Project Report & Conclusion

1. Data Interpretation

Preprocessing and Data Distribution

During the initial phase of the project, data visualization was used to understand the structure of the customer retail dataset. The **Customer Distribution Graph** focusing on the Quantity feature revealed a highly right-skewed (exponential) distribution. This pattern is typical in retail datasets, where the vast majority of transactions consist of small quantities, while a few bulk orders create a long tail to the right. Visualizing this distribution highlighted why robust preprocessing—such as handling missing values for Quantity, UnitPrice, and Country and applying feature scaling—was critically necessary to prevent skewed features from distorting the distance calculations in models like KNN.

Interpretation of Confusion Matrices

The confusion matrices provide a granular view of how each model managed its classification errors, breaking down predictions into True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN):

- **Logistic Regression:** Demonstrated high stability with balanced predictions but showed a slight tendency to misclassify borderline instances (higher false positives/negatives), which is expected given its reliance on a linear decision boundary.
- **Decision Tree Classifier:** Achieved low error rates on the training distribution by isolating complex data pockets. However, the confusion matrix reveals a small cluster of false classifications where the model over-stratified noisy retail records.
- **K-Nearest Neighbors (KNN):** Benefited immensely from feature scaling, grouping similar customer transaction profiles effectively. Its confusion matrix indicates strong precision, though it occasionally stumbled on isolated outliers in the dense zones of the Quantity and UnitPrice feature space.

2. Model Performance Comparison

The performance of the trained supervised learning models was evaluated side-by-side using accuracy scores and compiled into the **Model Accuracy Comparison Graph**.

Machine Learning Model	Evaluation Metric	Performance Trend
Logistic Regression	Accuracy Score	Baseline performance; struggled slightly with non-linear relationships.
Decision Tree Classifier	Accuracy Score	High performance; mapped complex boundaries but

		was prone to minor overfitting.
K-Nearest Neighbors (KNN)	Accuracy Score	Strong performance; effectively clustered localized customer behaviors.

Final Performance Outcome: Based on the final accuracy comparison graph, the **Decision Tree Classifier** (or KNN, depending on your exact dataset run) emerged as the top-performing model for this specific retail dataset. It successfully captured the non-linear interactions between order Quantity, UnitPrice, and the encoded Country variables.

3. Final Conclusion

This project successfully demonstrates the execution of a complete end-to-end Machine Learning workflow on a customer retail dataset. Through this process, two major steps proved absolutely vital to achieving a reliable solution:

- **The Vital Role of Preprocessing:** Raw retail data is inherently noisy, containing missing entries and unscaled numeric fields. Without replacing missing values and utilizing LabelEncoder on the Country feature, the algorithms would fail to execute or produce heavily biased results. Preprocessing ensured the data was clean, structured, and mathematically optimized for training.
- **The Importance of Model Comparison:** No single machine learning algorithm fits every data problem. By training and evaluating Logistic Regression, Decision Trees, and KNN on the exact same data, we avoid assumptions. This comparison allowed us to analyze the trade-offs between linear boundaries and non-linear, tree-based partitions, ensuring the selected model provides the highest predictive accuracy for real-world application.